

Rational Numbers on the Number Line

Lesson 9-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$1/2 = 0.5$, $-3/4 = -0.75$, $6 = 6/1$ — all rational numbers

Rational number

Any number you can write as a fraction. This includes fractions, decimals, and integers.

$3/4$ means 3 out of 4 equal parts — like 3 slices of a pizza cut into 4

Fraction

A number that shows part of a whole, like $3/4$.

$0.5 = 1/2$ (halfway between 0 and 1 on the number line)

Decimal

A number with a dot, like 0.5, that shows a part less than one.

A line with marks at -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2

Number line

A straight line with numbers spaced out evenly.

$1/2 = 0.5 = 2/4$ — all name the same point on the number line

Equivalent

Having the same value, just written a different way.

..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes

- Captain Vega's advanced treasure map uses precise coordinates — not just whole numbers!
- One clue says the treasure is at -2.5 on the map (halfway between -2 and -3).
- Another clue points to $\frac{3}{4}$ of the way from 0 to 1 .
- A third clue leads to $-1\frac{1}{2}$.
- Plot each rational number at its precise location on the number line.

Think About It

- Where would -2.5 fall on a number line between -3 and -2 ?
- How is plotting $\frac{3}{4}$ different from plotting a whole number?
- Can a negative number be a fraction or decimal?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

How can a fraction like $\frac{1}{2}$, a decimal like 0.75, and an integer like -3 all be rational numbers?

Level 1 support

Start here: All of these are rational numbers because each one can be ____.

A rational number is any number you can write as ____.

Un numero racional es cualquier numero que puedes escribir como ____.

For example, -3 can be written as ____.

Por ejemplo, -3 se puede escribir como ____.

Word bank: rational number fraction decimal integer number line

Listen for: Student explains a rational number is any number that can be written as a fraction, including integers and decimals.

Level 2

Push students to explain how to place $\frac{1}{2}$, 0.75, and -3 on the same number line.

- To place 0.75 on the number line, I would ____.
- I can compare a fraction and a decimal by ____.

EXPLORE

Where would you place $-1\frac{1}{2}$ on the number line, and how do you decide between which two integers it falls?

Level 1 support

Start here: $-1\frac{1}{2}$ falls between ____ and ____ because ____.

$-1\frac{1}{2}$ is between ____ and ____.

$-1\frac{1}{2}$ esta entre ____ y ____.

It is closer to ____ because ____.

Esta mas cerca de ____ porque ____.

Word bank: rational number fraction number line integer negative

Listen for: Student places $-1\frac{1}{2}$ between -1 and -2 , halfway, reasoning from the value of the fraction part.

Level 2

Ask students to compare $-1\frac{1}{2}$ and -1 using the number line and justify which is greater.

- -1 is greater than $-1\frac{1}{2}$ because ____.
- On the number line, ____ is to the right of ____.

CONNECT

How can rewriting a fraction as a decimal (or a decimal as a fraction) help you compare two rational numbers?

Level 1 support

Start here: Rewriting the numbers in the same form helps because ____.

If I turn $\frac{3}{4}$ into a decimal, I get ____.

Si convierto $\frac{3}{4}$ en decimal, obtengo ____.

Now I can compare them because ____.

Ahora puedo compararlos porque ____.

Word bank: rational number fraction decimal compare number line

Listen for: Student explains converting to a common form (both decimals or both fractions) makes comparison and ordering easier.

Level 2

Push students to generalize a strategy for ordering a mixed list of fractions, decimals, and integers.

- My strategy for ordering a mixed list is ____.
- Putting them all in the same form helps because ____.

REFLECT

Is every integer a rational number? Is every rational number an integer? Explain your thinking.

Level 1 support

Start here: Every integer is a rational number, but ____.

Every integer is rational because it can be written as ____.

Todo entero es racional porque se puede escribir como ____.

But not every rational number is an integer because ____.

Pero no todo numero racional es un entero porque ____.

Word bank: rational number integer fraction decimal number line

Listen for: Student explains integers are rational (write as a fraction over 1) but rationals like $\frac{1}{2}$ are not integers.

Level 2

Ask students to critique the statement 'all fractions are integers' with an example.

- The statement is false because ____.
- An example that disproves it is ____.

Guided Examples

Example 1

Which point is located between -1 and -2 on the number line?

Solution: -1.5 is between -1 and -2 because it is 0.5 units to the left of -1 (or 0.5 units to the right of -2).

Answer: B. -1.5

Example 2

Which fraction is equivalent to 0.25?

Solution: 0.25 means 25 hundredths = $25/100 = 1/4$.

Answer: A. $1/4$

Example 3

Where is $-3/4$ located on the number line?

Solution: $-3/4 = -0.75$, which is between 0 and -1. It is $3/4$ of the way from 0 toward -1.

Answer: A. Between 0 and -1

Write About the Math

The Writing Revolution

I can explain my reasoning using the words rational number, fraction, decimal, and number line.

1. Kernel Sentence subject + verb

Model: Rational number is any number you can write as a fraction. This includes fractions, decimals, and integers.

Número racional es cualquier número que puedes escribir como fracción. Incluye fracciones, decimales y enteros.

Write a kernel sentence about rational number. Use a subject and a verb.

Escribe una oración base sobre número racional. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Rational number matters in math

Número racional importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Rational number matters in math because ____.

Número racional importa en matemáticas porque ____.

but
pero

Rational number matters in math, but ____.

Número racional importa en matemáticas, pero ____.

so
entonces

Rational number matters in math, so ____.

Número racional importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement Afirmación

Tell one true fact about rational number.
Di un hecho verdadero sobre rational number.

Rational number ____.

Question Pregunta

Ask a question about rational number.
Haz una pregunta sobre rational number.

How does ____ ?

¿Cómo ____ ?

Exclamation Exclamación

Show excitement about rational number.
Muestra entusiasmo sobre rational number.

Wow, ____ !

¡Guau, ____ !

Command Mandato

Tell a partner what to do with rational number.
Dile a un compañero qué hacer con rational number.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

A rational number is any number you can write as ____.

Un numero racional es cualquier numero que puedes escribir como ____.

For example, -3 can be written as ____.

Por ejemplo, -3 se puede escribir como ____.

-1 1/2 is between ____ **and** ____.

-1 1/2 esta entre ____ *y* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Where is $-\frac{3}{4}$ located on the number line?

- A. Between 0 and -1
- B. Between -3 and -4
- C. Between -1 and -2
- D. Between 0 and 1

Show your work:

2. Which rational number is closest to 0 on the number line?

- A. $-\frac{1}{4}$
- B. 0.5
- C. -2
- D. 1.75

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Name three different rational numbers between -1 and 0. Write each as both a fraction and a decimal. Explain how you know they are between -1 and 0.

Sentence starter: Three rational numbers between -1 and 0 are: $\frac{\quad}{\quad} = \quad$, $\frac{\quad}{\quad} = \quad$, and $\frac{\quad}{\quad} = \quad$. I know they are between -1 and 0 because $\quad$.

Show your work:

Reflect — Exit Ticket

Which rational number is located between -2 and -3 on the number line?

- A. -1.5
- B. -2.75
- C. -3.5
- D. 2.5

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Between 0 and -1 — $-3/4 = -0.75$, which is between 0 and -1. It is $3/4$ of the way from 0 toward -1.
2. **Try It 2:** A. $-1/4$ — $|-1/4| = 0.25$, $|0.5| = 0.5$, $|-2| = 2$, $|1.75| = 1.75$. Since 0.25 is the smallest absolute value, $-1/4$ is closest to zero.
3. **Exit Ticket:** B. -2.75 — -2.75 is between -2 and -3 because it is 0.75 units to the left of -2. -1.5 is between -1 and -2, and -3.5 is between -3 and -4.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rational number is any number you can write as a fraction. This includes fractions, decimals, and integers.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).